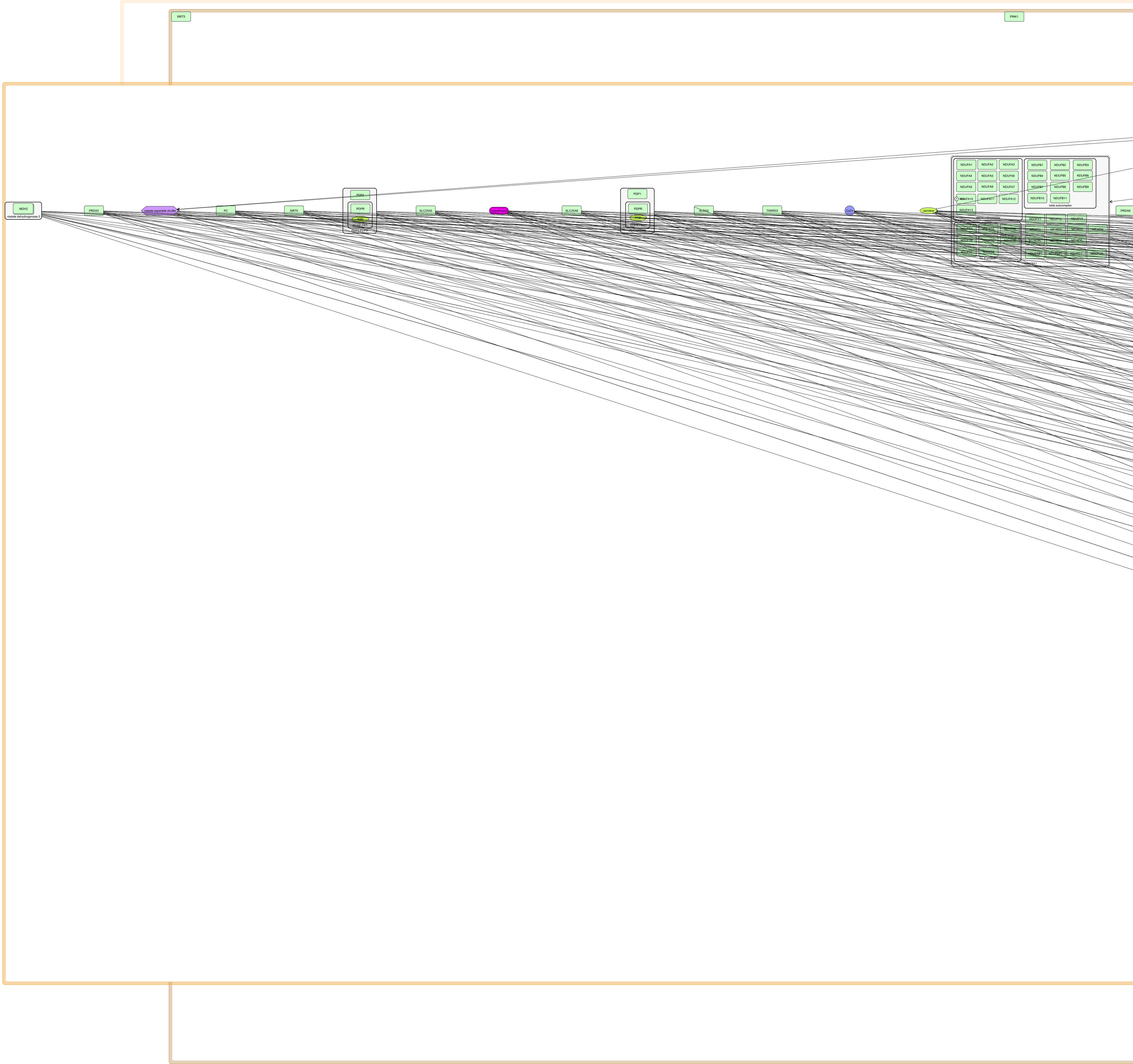
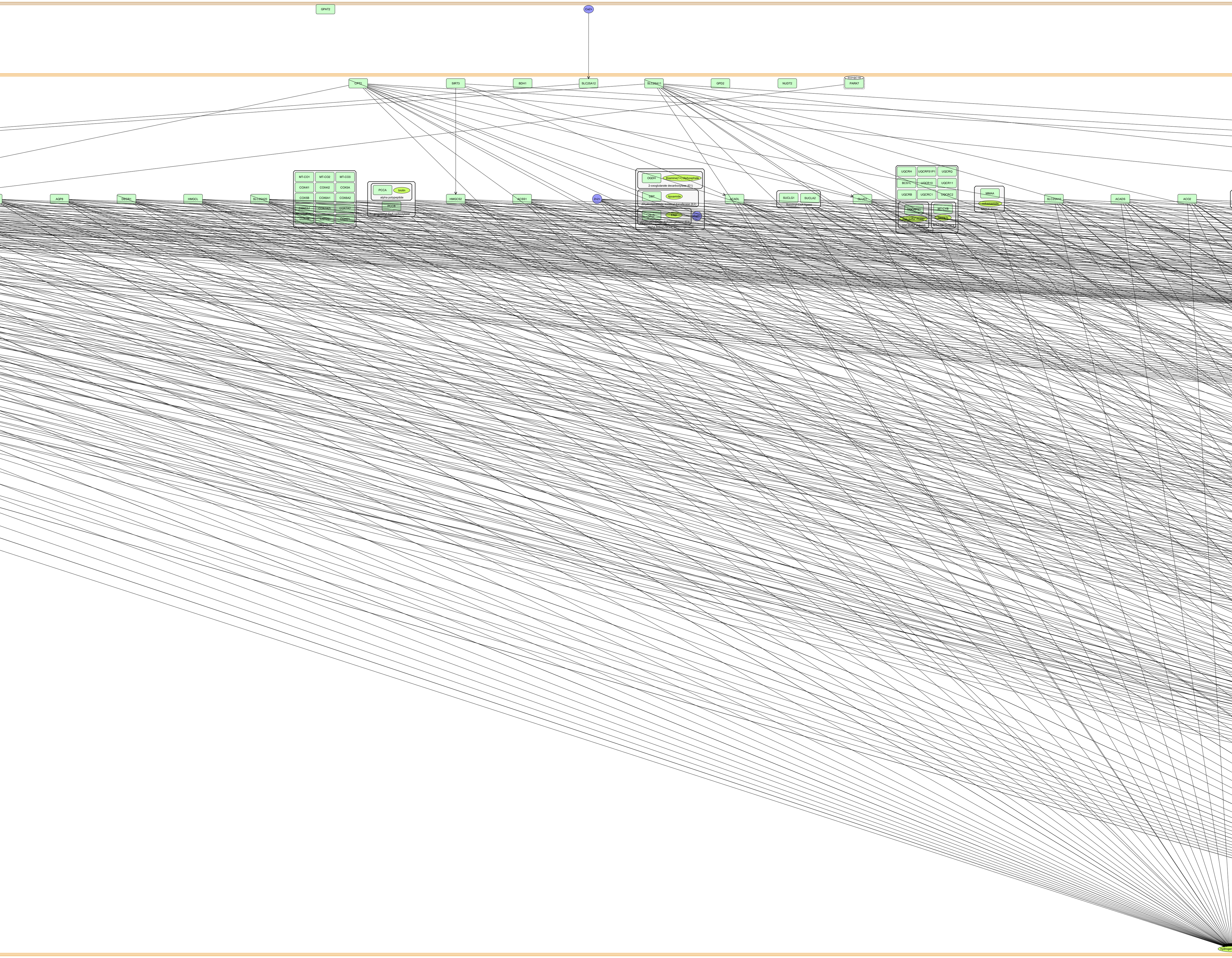
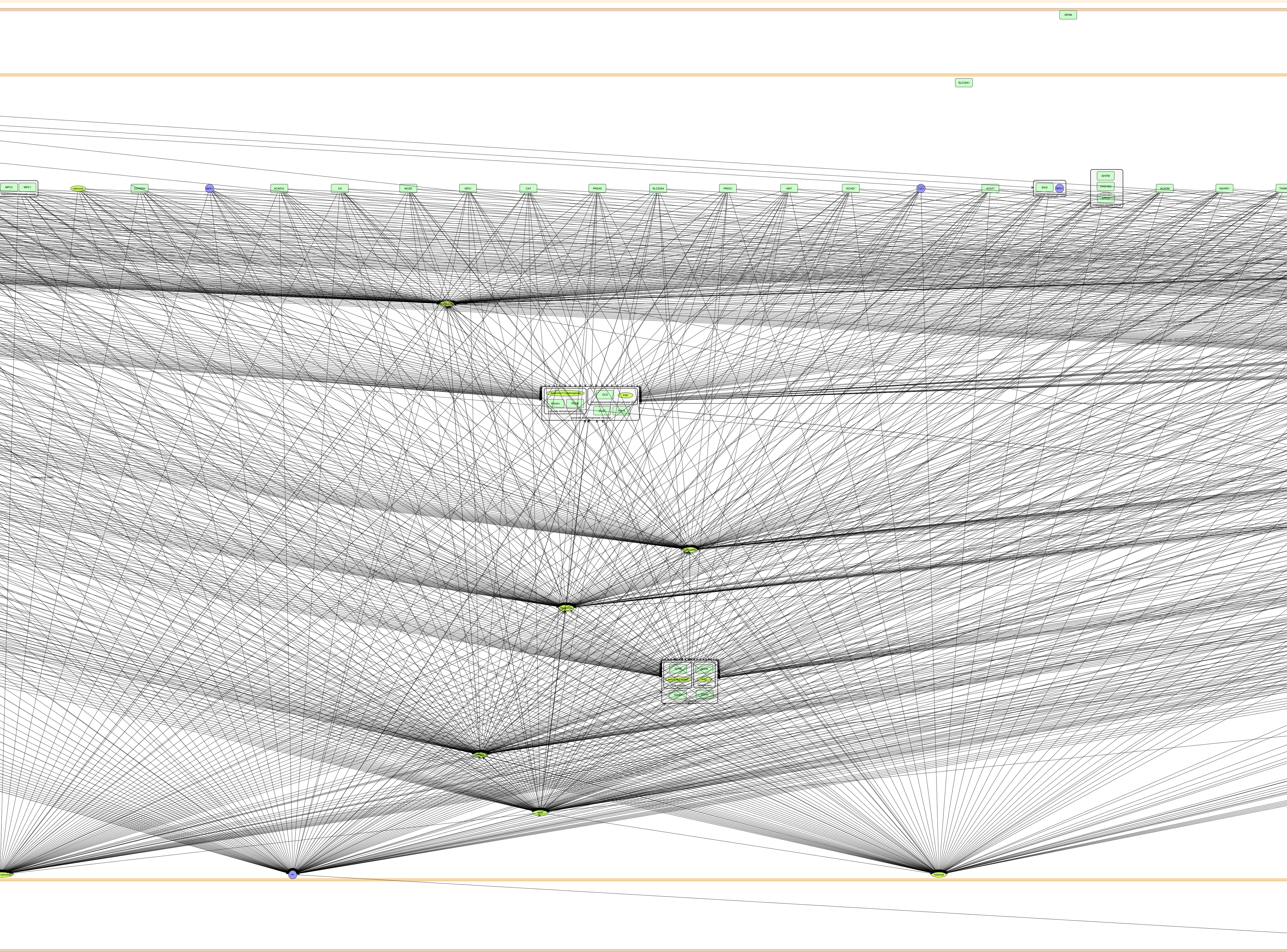
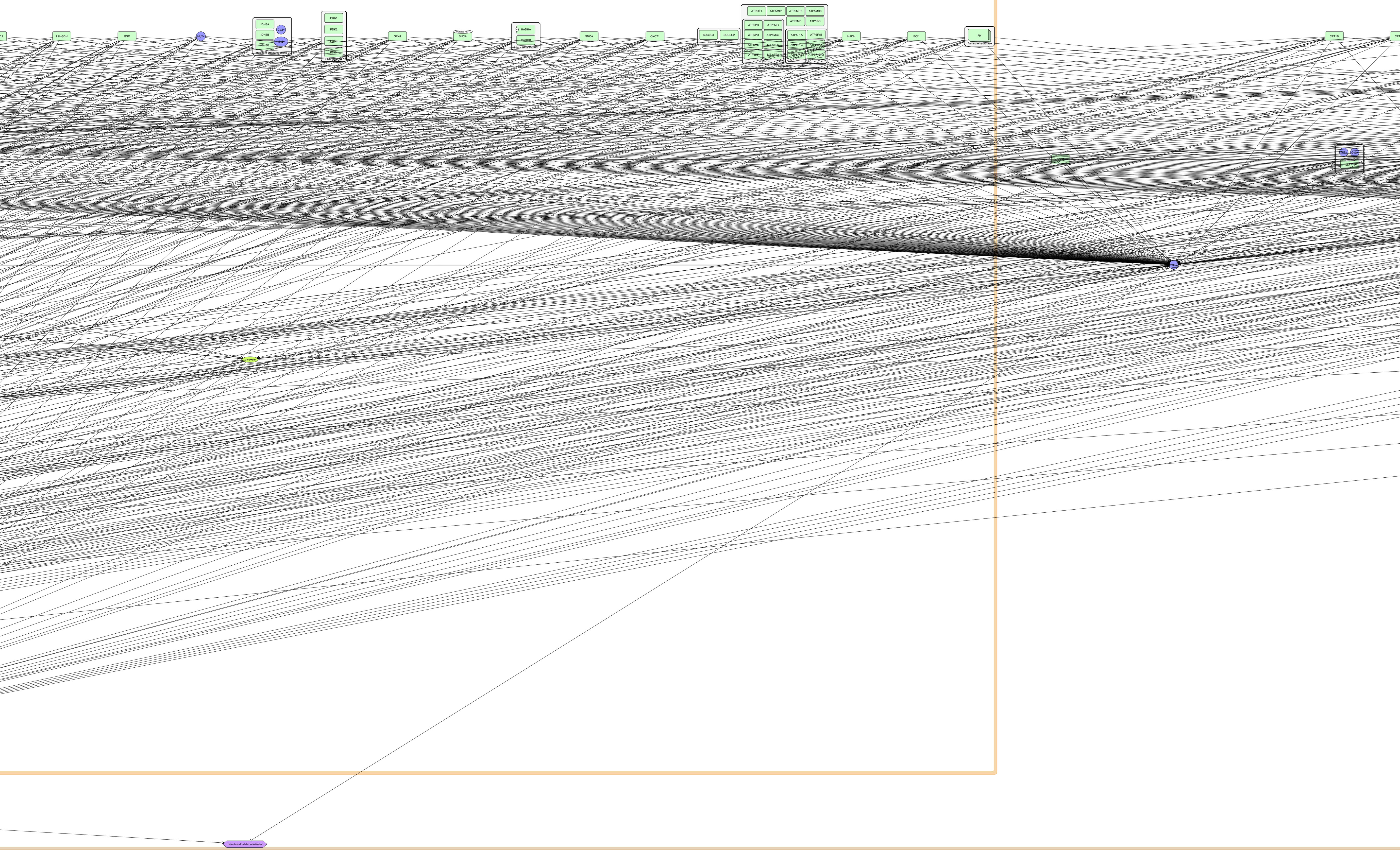
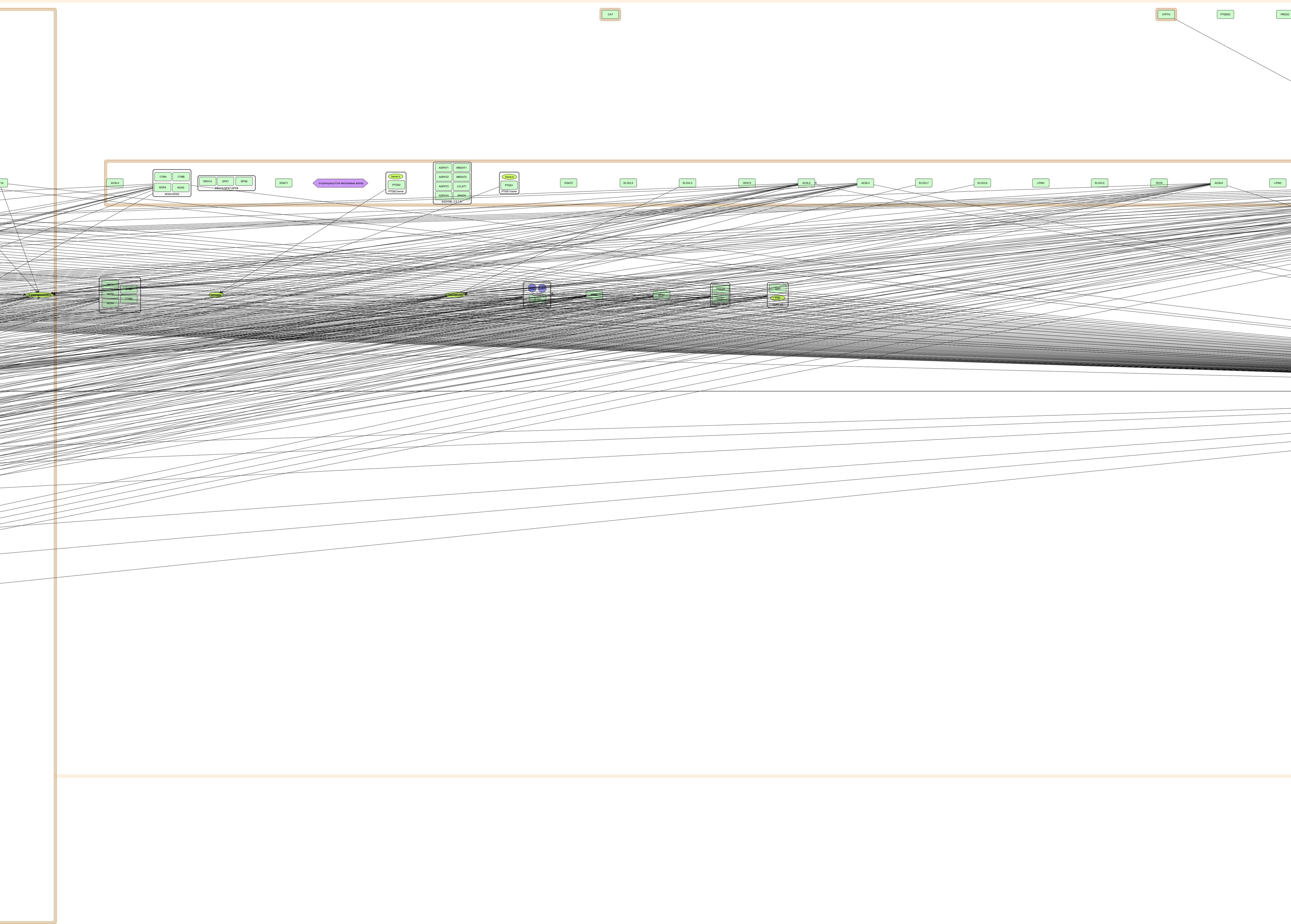
SDHB
SDHA
iron-sulfur cluster
FAD
Fe-S protein
flavoprotein
SDHC
SDHD
complex II
SIRT3
NDUFA9
NDUFA10
PINK1
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA9
NDUFA8
NDUFA7
NDUFA10
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
complex I
PARK7
ROS
mutated AS51
SNCA
SNCA
UQCRH
UQCRFS1P1
UQCRQ
SOD1
H+
mitochondrial depolarization
FH
fumarate hydratase
MDH2
malate dehydrogenase 2
H+
pyruvate
MPC2
MPC1
mitochondrial pyruvate carrier
(S)-malate(2-)
citrate(3-)
SLC25A1
SLC25A10
glutathione
glutathione
SLC25A11
NADH
NAD(+)
malate-aspartate shuttle
Ca2+
SLC25A12
NNT
H+
NADPH
SLC25A6
SLC25A4
SLC25A5
ADP
ATP
mitochondrial matrix
mitochondrion
neuron
OXCT1
acetyl-CoA
HMGCL
BDH1
MMAA
cobamamide
MMAA dimer
MCEE
PCCA
biotin
alpha polypeptide
PCCB
proprionyl-CoA carboxylase
acetoacetate
(R)-3-hydroxybutyrate
succinic acid
succinyl-CoA
(R)-methylmalonyl-CoA
(S)-methylmalonyl-CoA
propionyl-CoA
DECR1
GPD2
sn-glycerol 3-phosphate
glycerone phosphate
superoxide
H+
Zn2+
Cu2+
SOD3
SOD3:Cu2+:Zn2+
O2
hydrogen peroxide
glutathione
glutathione disulfide
H2O
GPX5
GPX6
GPX5:GPX6
GPX3
ATP7A
Golgi
ATOX1
ADP
phosphate(3-)
Cu1+
Zn2+
Zn2+
CCS
SOD1
CCS:Zn2+:Cu1+:SOD1:Zn2+
Zn2+
Cu2+
oxidized
SOD1
SOD1:Zn2+:Cu2+
TXNRD1
FAD
TXNRD1:FAD
H+
NADPH
oxidized
TXN
oxidized
TXN
hC TXN:hC TXN
H2O
PRDX1
PRDX2
PRDX5
H2O
nitrite
superoxide
NADP+
NADPH
GPX1
PRDX6
GSTP1
GPX2
NCF1
NCF2
NCF4
CYBA
CYBB
NOX2
phagocytic vesicle
CYBA
CYBB
NOX4
NOX5
NOX4:NOX5
hydrogen peroxide
H2O
P4HB
oxidized
P4HB
ERO1A
GPX7
GPX8
ERO1A:GPX7:GPX8
ER
CAT
hydrogen peroxide
H2O
peroxisome
ACACB
AMPK
ACACB
hydrogencarbonate
acetyl-CoA
ACACA
malonyl-CoA
ACSL6
ACSL5
ACSL3
ACSL1
ELOVL7
ELOVL6
ELOVL3
CPT1B
malonyl-CoA
palmitoyl-CoA
3-oxooctadecanoyl-CoA
HSD17B12
3-hydroxyoctadecanoyl-CoA
3-hydroxyacyl-CoA dehydratase activity
trans-2-octadecenoyl-CoA
TECR
glycerone phosphate
stearoyl-CoA
GPD1
glycerol
GK
sn-glycerol 3-phosphate
GPAT4
GPAT3
GPAM
GPAT2
AGPAT1
MBOAT1
AGPAT2
MBOAT2
AGPAT3
LCLAT1
AGPAT4
ABHD5
ENZYME: 2.3.1.51
1-acyl-sn-glycerol 3-phosphate
1,2-diacyl-sn-glycerol 3-phosphate
LPIN1
LPIN2
LPIN3
fatty acyl-CoA
1,2-diacyl-sn-glycerol
DGAT1
DGAT2
coenzyme A
triglyceride
ER
prostaglandin E2
PTGES2
PTGES
PTGES3
prostaglandin D2
PTGDS
prostaglandin F2 alpha
prostaglandin H2
AKR1C3
peroxide
ROS
heme b
PTGS1
PTGS1:heme
heme b
PTGS2
PTGS2:heme
prostaglandin G2
Arachidonic acid
ACSL4
arachidonoyl-CoA
ELOVL5
3ODCT-CoA
icosanoyl-CoA
ELOVL7
3-oxodocosanoyl-CoA
ELOVL4
tetracosanoyl-CoA
ER

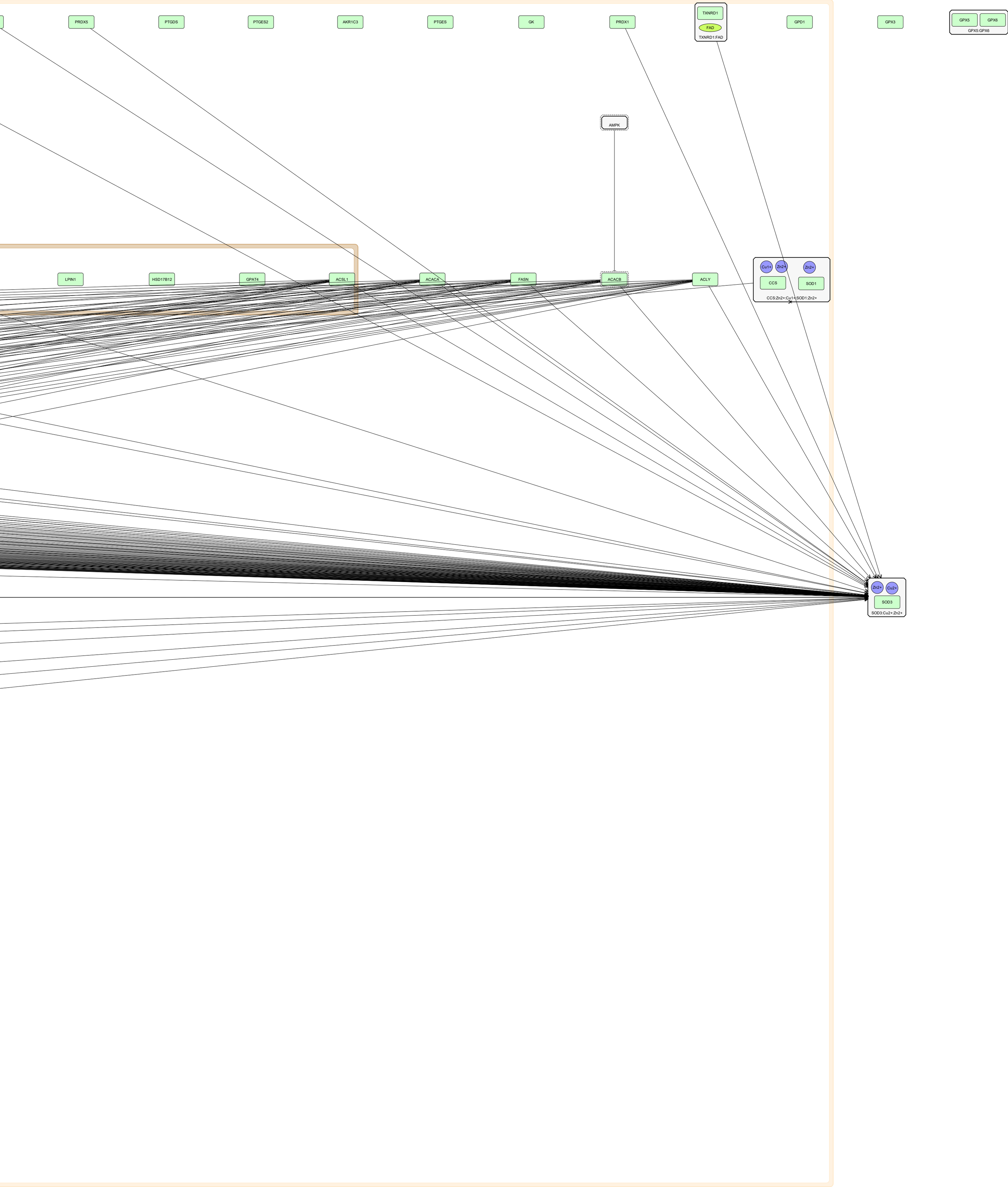

SIRT3
PINK1
MDH2
malate dehydrogenase 2
PRDX2
malate-aspartate shuttle
PC
SIRT5
PDP2
PDPR
FAD
PDPR:FAD
PDP2:PDPR
SLC25A5
myxothiazol
SLC25A6
PDP1
PDPR
FAD
PDPR:FAD
PDP1:PDPR
ALDH2
TXNRD2
Co2+
carnitine
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA9
NDUFA8
NDUFA7
P
ND1
NDUFA10
NDUFA11
NDUFA12
NDUFA13
alpha subcomplex
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta subcomplex
NDUFS1
NDUFS2
NDUFS3
NDUFS4
NDUFS5
NDUFS6
NDUFS7
NDUFS8
Fe-S protein
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
NDUFAB1
NDUFAF1
NDUFC1
NDUFC2
complex I
PRDX6


GPAT2
Ca2+
CPT2
SIRT3
BDH1
SLC25A12
SLC25A11
GPD2
NUDT2
SO2H@C106
PARK7
AQP8
DECR1
HMGCL
SLC25A20
MT-CO1
MT-CO2
MT-CO3
COX4I1
COX4I2
COX5A
COX5B
COX6A1
COX6A2
COX7A1
COX7A2L
COX7A2
COX7B
COX7C
COX8A
complex IV
PCCA
biotin
alpha polypeptide
PCCB
proprionyl-CoA carboxylase
HMGCS2
ACSS1
Zn2+
OGDH
thiamine(1+) diphosphate
2-oxoglutarate decarboxylase (E1)
DBT
lipoamide
dihydrolipoamide succinyl-transferase (E2)
DLD
FAD
Ca2+
dihydrolipoamide dehydrogenase (E3)
alpha-ketoglutarate dehydrogenase
ACADL
SUCLG1
SUCLA2
Succinyl-CoA ligase
GLUD1
UQCRH
UQCRFS1P1
UQCRQ
BCS1L
UQCR10
UQCR11
UQCRB
UQCRC1
UQCRC2
UQCRFS1
iron-sulfur cluster
iron-sulfur subunit
MT-CYB
heme b
MT-CYB heme b
complex III
MMAA
cobamamide
MMAA dimer
SLC25A10
ACADS
ACO2
hydrogen


GPAM
SLC25A1
MPC2
MPC1
mitochondrial pyruvate carrier
rotenone
D2HGDH
Mn2+
ACADVL
CS
MCEE
GPX1
CAT
PRDX5
SLC25A4
PRDX1
NNT
ECHS1
Ca2+
ACAT1
IDH2
Mn2+
isocitrate dehydrogenase
GHITM
CHCHD2
reduced
CYCS
CHCHD2:GHITM:CYCS
ACADM
ADHFE1
TXNR
acetyl-CoA
thiamine(1+) diphosphate
PDHA1
PDHB
pyruvate dehydrogenase (E1)
DLD
FAD
dihydrolipoamide dehydrogenase (E3)
DLAT
PDHX
pyruvate dehydrogenase complex
mitochondrial matrix
superoxide
ubiquinol
SDHB
iron-sulfur cluster
Fe-S protein
SDHA
FAD
flavoprotein
SDHC
SDHD
complex II
NADH
ADP
n peroxide
H+
hydroxyl


L2HGDH
GSR
Mg2+
IDH3A
IDH3B
IDH3G
Ca2+
Mn2+
isocitrate dehydrogenase 3
PDK1
PDK2
PDK3
PDK4
PDK isoforms
GPX4
mutated A53T
SNCA
Ac
HADHA
HADHB
trifunctional Protein
SNCA
OXCT1
SUCLG1
SUCLG2
Succinyl-CoA ligase
ATP5IF1
ATP5MC1
ATP5MC2
ATP5MC3
ATP5MF
ATP5PO
ATP5PB
ATP5MG
ATP5PD
ATP5MGL
ATP5ME
MT-ATP6
ATP5PF
MT-ATP8
F(0)
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5F1EP2
F(1)
complex V
HADH
ECI1
FH
fumarate hydratase
CPT1B
SOD1
Zn2+
Cu2+
SOD1
SOD1-Zn2+-Cu2+
H+
pyruvate
mitochondrial depolarization


CAT
ATP7A
PTGES3
PRDX2
ACSL4
CYBA
CYBB
NOX4
NOX5
NOX4:NOX5
ERO1A
GPX7
GPX8
ERO1A:GPX7:GPX8
DGAT1
3-hydroxyacyl-CoA dehydratase activity
heme b
PTGS2
PTGS2:heme
AGPAT1
MBOAT1
AGPAT2
MBOAT2
AGPAT3
LCLAT1
AGPAT4
ABHD5
ENZYME: 2.3.1.51
heme b
PTGS1
PTGS1:heme
DGAT2
ELOVL4
ELOVL3
GPAT3
ACSL5
ACSL3
ELOVL7
ELOVL6
LPIN2
ELOVL5
TECR
ACSL6
LPIN3
L-palmitoylcarnitine
NCF1
NCF2
NCF4
CYBA
CYBB
NOX2
peroxide
malonyl-CoA
Zn2+
Cu2+
oxidized
SOD1
SOD1:Zn2+:Cu2+
GPX1
GPX2
PRDX6
GSTP1
PRDX6:GSTP1
GSR
FAD
GSR:FAD


PRDX5
PTGDS
PTGES2
AKR1C3
PTGES
GK
PRDX1
TXNRD1
FAD
TXNRD1:FAD
GPD1
GPX3
GPX5
GPX6
GPX5:GPX6
AMPK
LPIN1
HSD17B12
GPAT4
ACSL1
ACACA
FASN
ACACB
ACLY
Cu1+
Zn2+
Zn2+
CCS
SOD1
CCS:Zn2+:Cu1+:SOD1:Zn2+
Zn2+
Cu2+
SOD3
SOD3:Cu2+:Zn2+